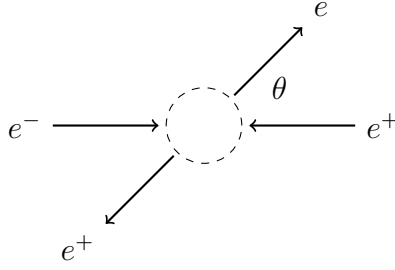


Bhabha scattering 1

Bhabha scattering is the process $e^- + e^+ \rightarrow e^- + e^+$.



Momentum vectors where $E = \sqrt{p^2 + m^2}$.

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}_{e^- \rightarrow} \quad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}_{\leftarrow e^+} \quad p_3 = \begin{pmatrix} E \\ p \sin \theta \cos \phi \\ p \sin \theta \sin \phi \\ p \cos \theta \end{pmatrix}_{e^- \nearrow} \quad p_4 = \begin{pmatrix} E \\ -p \sin \theta \cos \phi \\ -p \sin \theta \sin \phi \\ -p \cos \theta \end{pmatrix}_{\nwarrow e^+}$$

Spinors for p_1 .

$$u_{11} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ -p \\ 0 \end{pmatrix}_{\text{spin up}} \quad u_{12} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ 0 \\ p \end{pmatrix}_{\text{spin down}}$$

Spinors for p_2 .

$$v_{21} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} p \\ 0 \\ E+m \\ 0 \end{pmatrix}_{\text{spin up}} \quad v_{22} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ -p \\ 0 \\ E+m \end{pmatrix}_{\text{spin down}}$$

Spinors for p_3 .

$$u_{31} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p_{3z} \\ p_{3x} + ip_{3y} \end{pmatrix}_{\text{spin up}} \quad u_{32} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ p_{3x} - ip_{3y} \\ -p_{3z} \end{pmatrix}_{\text{spin down}}$$

Spinors for p_4 .

$$v_{41} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} p_{4z} \\ p_{4x} + ip_{4y} \\ E+m \\ 0 \end{pmatrix}_{\text{spin up}} \quad v_{42} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} p_{4x} - ip_{4y} \\ -p_{4z} \\ 0 \\ E+m \end{pmatrix}_{\text{spin down}}$$

The scattering amplitude $\mathcal{M}$ is

$$\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$$

where $\mathcal{M}_1$ is the scattering amplitude

$$\mathcal{M}_1 = \frac{\alpha}{t} (\bar{v}_{2b} \gamma^\mu v_{4d}) (\bar{u}_{3c} \gamma_\mu u_{1a})$$

and $\mathcal{M}_2$ is the annihilation amplitude

$$\mathcal{M}_2 = \frac{\alpha}{s} (\bar{u}_{3c} \gamma^\mu v_{4d}) (\bar{v}_{2b} \gamma_\mu u_{1a})$$

The expected density $\langle |\mathcal{M}|^2 \rangle$ is the average over spin states.

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{4} \sum_{abcd} |\mathcal{M}|^2$$

Expand the summand.

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{4} \sum_{abcd} (\mathcal{M}_1 \mathcal{M}_1^* - \mathcal{M}_1 \mathcal{M}_2^* - \mathcal{M}_2 \mathcal{M}_1^* + \mathcal{M}_2 \mathcal{M}_2^*)$$

The summation results are

$$\begin{aligned} \sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* &= \frac{\alpha^2}{t^2} (8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4) \\ \sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* &= \frac{\alpha^2}{st} (-8u^2 + 64m^2u - 96m^4) \\ \sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* &= \frac{\alpha^2}{s^2} (8t^2 + 8u^2 - 64m^2t - 64m^2u + 192m^4) \end{aligned}$$

where s , t , and u are the Mandelstam variables

$$\begin{aligned} s &= (p_1 + p_2)^2 \\ t &= (p_1 - p_3)^2 \\ u &= (p_1 - p_4)^2 \end{aligned}$$

For clarity define

$$\begin{aligned} f_{11} &= 8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4 \\ f_{12} &= -8u^2 + 64m^2u - 96m^4 \\ f_{22} &= 8t^2 + 8u^2 - 64m^2t - 64m^2u + 192m^4 \end{aligned}$$

Now write $\langle |\mathcal{M}|^2 \rangle$ more compactly as

$$\langle |\mathcal{M}|^2 \rangle = \frac{\alpha^2}{4} \left(\frac{f_{11}}{t^2} - \frac{2f_{12}}{st} + \frac{f_{22}}{s^2} \right)$$

For $E \gg m$ a useful approximation is to set $m = 0$ and obtain

$$\begin{aligned}f_{11} &= 8s^2 + 8u^2 \\f_{12} &= -8u^2 \\f_{22} &= 8t^2 + 8u^2\end{aligned}$$

Then $\langle |\mathcal{M}|^2 \rangle$ simplifies as

$$\langle |\mathcal{M}|^2 \rangle = 2\alpha^2 \left(\frac{s^2 + u^2}{t^2} + \frac{2u^2}{st} + \frac{t^2 + u^2}{s^2} \right)$$

For $m = 0$ the Mandelstam variables are

$$\begin{aligned}s &= 4E^2 \\t &= -2E^2(1 - \cos \theta) \\u &= -2E^2(1 + \cos \theta)\end{aligned}$$

Hence

$$\langle |\mathcal{M}|^2 \rangle = 2\alpha^2 \left(\frac{1 + \cos^4(\theta/2)}{\sin^4(\theta/2)} - \frac{2 \cos^4(\theta/2)}{\sin^2(\theta/2)} + \frac{1 + \cos^2 \theta}{2} \right)$$

This can be written more compactly as

$$\langle |\mathcal{M}|^2 \rangle = \alpha^2 \left(\frac{\cos^2 \theta + 3}{\cos \theta - 1} \right)^2$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\langle |\mathcal{M}|^2 \rangle}{4s} = \frac{\alpha^2}{16E^2} \left(\frac{\cos^2 \theta + 3}{\cos \theta - 1} \right)^2$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{16E^2} \left(\frac{\cos^2 \theta + 3}{\cos \theta - 1} \right)^2 (\hbar c)^2$$

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